

Introduction

❑ **Welding:** In general, it is a process of joining two material plates and make an integrated one.

- ❖ The large bulk of materials that are welded are **metals and their alloys**. The welding is also applied to the joining of other material such as **thermoplastics**.
- ❖ In welding **heat is supplied** either by **electrical arc** or by a **gas torch** or by **some other means**.
- ❖ The most essential requirement is ***Heat*** but in some processes ***Pressure*** is also employed.

History of Welding

❑ Middle Ages:

Blacksmiths of the Middle Ages welded various types of iron tools by hammering. The welding methods remained more or less unchanged until the dawn of the 19th century.

❑ Late 19th Century

- Engineers/scientists apply advances of electricity to heat and join metals (Joule, Le Chatelier, etc.)

❑ Early 20th Century

- Prior to 1st World War welding was not trusted as a method to join two metals due to **crack issues**.

❑ 1930's & 40's

- Industrial welding gains acceptance and is **used extensively in the war effort** to build tanks, aircraft, ships etc. The use of welding in today's technology is **extensive**. It is **a remarkable rise** since about 1930.

History of Welding

❑ **19th Century (1800):** In this century, major weldings were made.

1830:

Englishman Edmund Davy discovered *acetylene in 1836* and acetylene was soon utilized by the welding industry.

1880:

In 1881, French scientist **Auguste De Meritens** succeeded in **fusing lead plates** by using the heat generated from an arc.

1890:

During the 1890's, one of the most popular welding methods was invented i.e. *carbon arc welding*. In this time, **thermite welding** was also invented in 1893.

History of Welding (*cont.*)

❑ 20th Century (1900):

1900:

- ✓ *Coated metal electrode* was first introduced by Strohmenger. A coating helped the arc to be much more stable.
- ✓ A number of other welding processes were developed during this period i.e. **seam welding, spot welding, flash butt welding, and projection welding.**

1919:

- ✓ After the end of World War I, the American Welding Society was established by **Comfort Avery Adams**. The aim of the society was the advancement of welding processes.

1920:

- ✓ Automatic welding was first introduced which was invented by **P. O. Nobel**. In 1920, an early **predecessor of GMAW** was invented by P. O. Nobel of General Electric.

History of Welding (*cont.*)

1930: The New York Navy Yard developed **stud welding**. Stud welding was increasingly used for the construction industry and also for shipbuilding.

1940: The **GTAW** was another significant milestone in the history of welding which was developed in **Battelle Memorial Institute in 1948**.

1960: There were several advancements in the welding industry during the 1960's. **Electroslag welding and Plasma arc welding** were invented during this time.

1990: In 1991, Welding Institute invented **FSW**. It is a solid state joining process which utilizes frictional heat of a rotating tool and stirring effect of the tool probe for solid state joining.

- The use of welding in today's technology is **extensive**. This growth is faster than the general industrial growth.

Common Welding Base Material

❑ Metals can be classified as:

1. Ferrous
2. Non-ferrous Material

1. Ferrous materials finding day-to-day welding application are:

- i) Wrought Iron (Less than 0.035% Carbon)
- ii) Cast Iron [Carbon and Silicon % are: **2.3 to 4.5%** and **0.5 to 3%** respectively]
- iii) Carbon Steel [Low (0.05–0.3%), Medium (0.30–0.59%) and High (0.6–1.5%)]
- iv) Cast Steels [Carbon content between **0.2 to 2.1%** by weight, depending on the grade, also other alloying elements manganese, chromium, vanadium, and tungsten]
- v) Stainless steel [More than **11.5%** chromium], etc.

Common Welding Base Material *cont.*

2. Non-Ferrous materials finding day-to-day welding application are:

- i) Aluminium and its alloys
- ii) Copper and its alloys
- iii) Magnesium and its alloys
- iv) Nickel and its alloys
- v) Zinc and its alloys,

General Advantage of Welding

□ Advantages:

- ❖ A good weld is **as strong as the base metal**.
- ❖ General welding equipment is **not very costly**.
- ❖ Portable welding equipment is also available.
- ❖ Welding **permits considerable freedom** in design.
- ❖ A large number of metals/ alloys both similar and dissimilar can be joined by welding.
- ❖ Welding can join workpieces by spots, as continuous pressure tight seams, end-to-end and in a number of other configurations.
- ❖ Welding can be **mechanized**.

General Disadvantage of Welding

❑ Disadvantages:

- ❖ Welding gives out harmful radiations (light), fumes and spatter.
- ❖ Welding results in residual stresses and distortion of the workpieces.
- ❖ Jigs and fixtures are generally required to hold and position the parts to be welded.
- ❖ A skilled welder is a must to produce a good welding job.
- ❖ Welding heat produces metallurgical changes. The structure of the welded joint is not same as that of the parent metal.
- ❖ A welded joint, for many reasons, needs stress-relief heat-treatment.

Welding as compared to casting

- ❑ Machine tool beds which were earlier cast are now fabricated using welding. In many fields welding has replaced casting processes.
- ❖ Some of the reasons for the same as follow:
 - ✓ Welding is **more economical** and is a **much faster process** as compared to casting.
 - ✓ Fabricated mild steel structures are **lighter** as compared to cast ones.
 - ✓ Fabricated mild steel structures have **more tensile strength** and **rigidity** as compared to cast ones.
 - ✓ Cost of **pattern** and **storing** is eliminated.

Welding as compared to casting *cont.*

- ✓ As compared to casting fewer persons are involved in a welding fabrication.
- ✓ **Structural shapes not easily obtainable** with casting can be produced by welding without much difficulty.
- ✓ Welding design involves **low costs and it is very flexible** also.
- ✓ Fabrication by **welding saves machining cost** involved in cast part.

Welding as compared to riveting

❑ Bridges, ships and boilers which were previously riveted are now welded.

- ❖ Welding is more economical and is a much faster process as compared to riveting.
- ❖ Welded pressure vessels are more pressure tight as compared to riveted ones.
- ❖ Ratio between weight of weld metal and the entire weight of structure is **much lesser** than the ratio between the weight of rivets and entire weight of the structure.
- ❖ Cover plates, connecting angles, gusset plates, etc., needed in riveted construction are not required when welding the structures.

Welding as compared to riveting *cont.*

- ❖ Members of such shape that present difficulty for riveting can be easily welded.
- ❖ Welding can be carried out at any point on a structure, but, riveting always requires **enough clearance to be done**.
- ❖ A welded structure possesses a better finish and appearance than the corresponding riveted structure.
- ❖ Layout for punching and drilling of holes is not required in welding.

Welding as compared to riveting *cont.*

- ❖ Drilling holes in the plate in order to accommodate rivets, breaks material continuity and weakens a riveted structure.
- ❖ Making changes in an already cast or riveted structure is extremely difficult, if not impossible. On the other hand a welded structure can be repaired without much difficulty.
- ❖ Welding can produce a 100% efficient joint which is difficult to make by riveting.
- ❖ Riveting high strength steels presents the problems of acquiring high strength steels rivets.

Practical Applications of Welding

❑ Welding has been employed in industry as a tool for:

- ❖ Regular fabrication** of automobile cars, air-crafts, refrigerators, ships, offshore structure etc.
- ❖ Repair and maintenance work**, e.g., joining broken parts, rebuilding worn out components, etc.

❑ A few important applications of welding are listed below:

i. Aircraft construction :

- (a) Welded engine mounts.
- (b) Turbine frame for jet engine.
- (c) Rocket motor fuel tanks.
- (d) Fittings, etc.

ii. Automobile construction :

- (a) Are welded car wheels.
- (b) Frame side rails.
- (c) Automobile frame, brackets etc.

Practical Applications of Welding (*cont.*)

iii. Bridges:

- (a) Pier construction.
- (b) Section lengths etc.

iv. Buildings:

- (a) Column base plates.
- (b) Trusses.
- (c) Erection of structure, etc.

v. Pressure vessels and tanks:

- (a) Clad and lined steel plates.
- (b) Shell construction.
- (c) Joining of nozzles to the shell, etc.

vi. Storage tanks:

- (a) Oil, gas and water storage tanks.

vii. Rail road equipment:

- (a) Rail
- (b) Under frame
- (c) Air receiver
- (d) Engine etc.

Practical Applications of Welding (*cont.*)

viii. Piping and pipelines:

- (a) Rolled plate pipings.
- (b) Open pipe joints.
- (c) Oil, gas and gasoline pipe lines, etc.

ix. Ships:

- (a) Shell frames.
- (b) Deck and bulkhead stiffeners.
- (c) Girders to shells etc.

x. Trucks and trailers.

xi. Machines tool frames, cutting tools and dies.

xii. Household and office furniture.

Practical Applications of Welding (*cont.*)

- ❑ **In addition, arc welding finds following applications in repair and maintenance work:**
- ❖ Repair of broken & damaged components and machinery such as tools, punches, dies, gears, press and machine tools frames.
- ❖ Fabrication of jigs, fixtures, clamps and other work holding devices.
- ❖ Being noiseless as compared to riveting, welding find extensive use, when making modifications, addition or extension in hospital buildings.

Classification of Welding Process

Joining Process

❑ There are basically two types of joining process:

1. Mechanical Bonding
2. Atomic Bonding

1. Mechanical Bonding classification:

- a) Temporary (With Screw Elements)
- b) Permanent/ Semi-permanent
 - i) Rivets
 - ii) Stitches
 - iii) Staples
 - iv) Shrink-fit

2. Atomic Bonding classification: Welding

Classification of Welding

❑ **Welding classification can be done based on the following aspects:**

1. Depending upon the source of heat
2. Depending upon the application of pressure
3. Depending upon the different phases of base and filler material
4. Depending upon the composition of the joint
5. Depending upon the position of electrode
6. Depending upon the mechanism

Types of Welding

□ Depending upon the source of heat:

- (i). Arc welding
- (ii). Gas welding
- (iii). Resistance welding
- (iv) Thermo-chemical welding process
- (v) Mechanical energy welding process
- (vi) Radiant energy welding process

Types of Welding

□ Different welding techniques name (depending on source of heat):

(i). Arc welding

- Carbon arc (CAW)
- Metal arc (SMAW)
- Tungsten inert gas(TIG/GTAW)
- Metal inert gas (MIG/GMAW)
- Plasma arc (PAW)
- Submerged arc (SAW)
- Electro-slag (ESW)
- Electro gas(ESW)

(ii). Gas Welding

- Oxy-acetylene
- Air-acetylene
- Oxy-hydrogen
- Pressure gas

(iii). Resistance Welding

- Butt
- Spot
- Seam

- Projection
- Percussion
- Flash Butt

(iv) Thermo-chemical welding process

- Thermit welding
- Atomic hydrogen welding

(v) Mechanical energy welding process

- Friction
- Ultrasonic
- Diffusion
- Forge
- Roll
- Explosive

(vi) Radiant energy welding process

- Electron-beam (EBW)
- Laser (LBM)

Types of Welding (*cont.*)

□ Depending upon the application of pressure (2-categories):

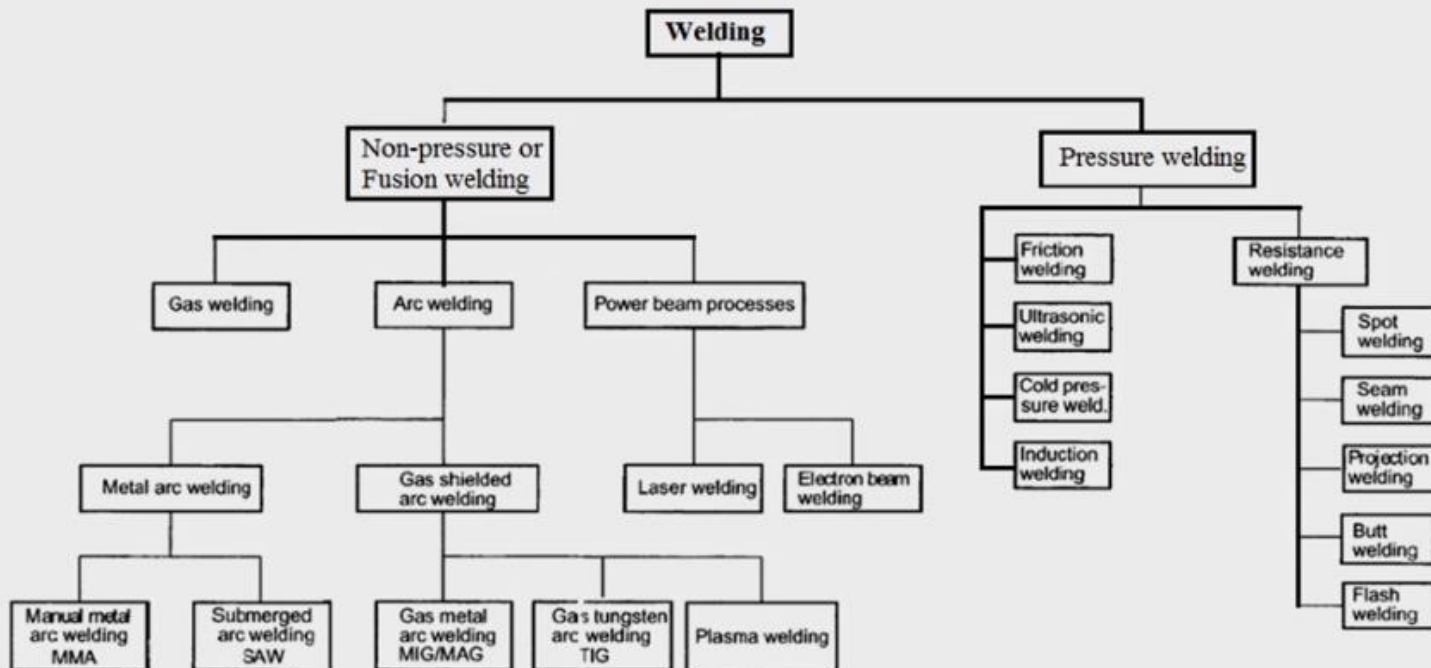
- **Pressure welding**

The pieces of metal to be joined are heated to a plastic state and forced together by external pressure. This is also known as **Plastic Welding**.

- **Non-Pressure welding or Fusion welding**

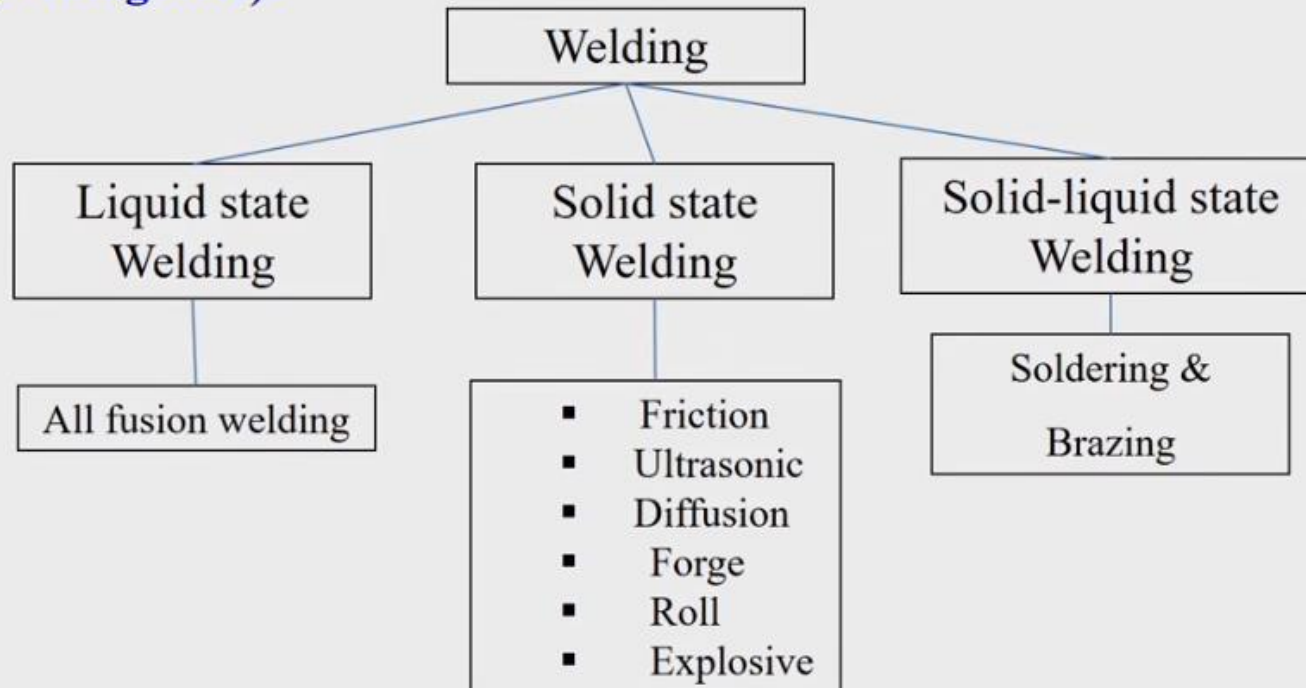
The material at the joint is heated to a molten state and allowed to solidify.

Types of Welding (Cont.)



Types of Welding (*cont.*)

- Depending upon the different phases of base and filler material (3-categories):



Types of Welding (*Cont.*)

❑ Depending upon the composition of the joint (3-categories):

1. Autogeneous
2. Homogeneous
3. Heterogeneous

1. **Autogeneous welding:** No filler material is added during this joining.

Ex.: All type of solid phase welding, resistance welding and non-consumable welding.

2. **Homogeneous welding:** The composition of filler material used during this joining is same as the parent material.

Ex.: Arc, Gas and Thermit welding.

3. **Heterogeneous welding:** The composition of filler material used during this joining is different as the parent material.

Ex.: Soldering and Brazing.

Types of Welding (Cont.)

□ Depending upon the position of welding (5-categories):

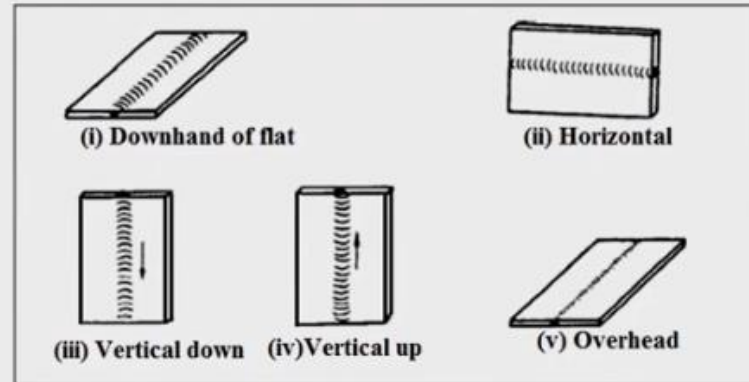
i) Downhand or flat

ii) Horizontal

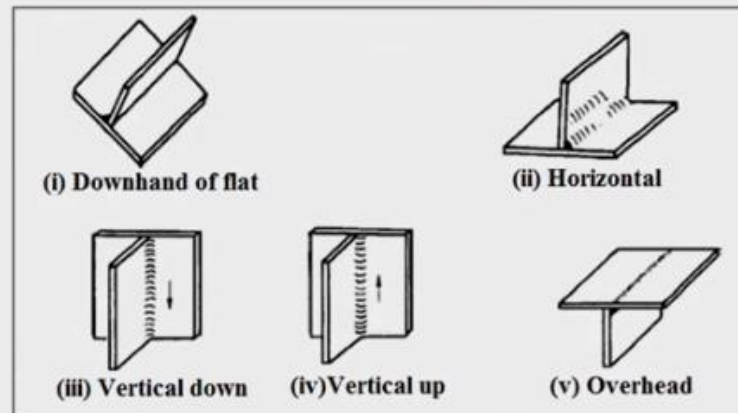
iii) Vertical down

iv) Vertical up

v) Overhead



(a) For Butt welding



(b) For Fillet welding

Types of Welding (*Cont.*)

❑ Depending upon the mechanism (3-categories):

1. **Manual Welding:**

In manual welding both **feeding of filler material and welding speed** are controlled manually. Example: SMAW.

2. **Semi-automatic:**

In this welding either feeding of filler material or welding speed is controlled automatically. Example: MIG, TIG etc.

3. **Fully automatic:**

In this welding both feeding of filler material and welding speed is controlled automatically. Example: SAW, Laser, EB etc

Type of Weld Joints

Type of Weld Joints

□ There are **five basic** joints:

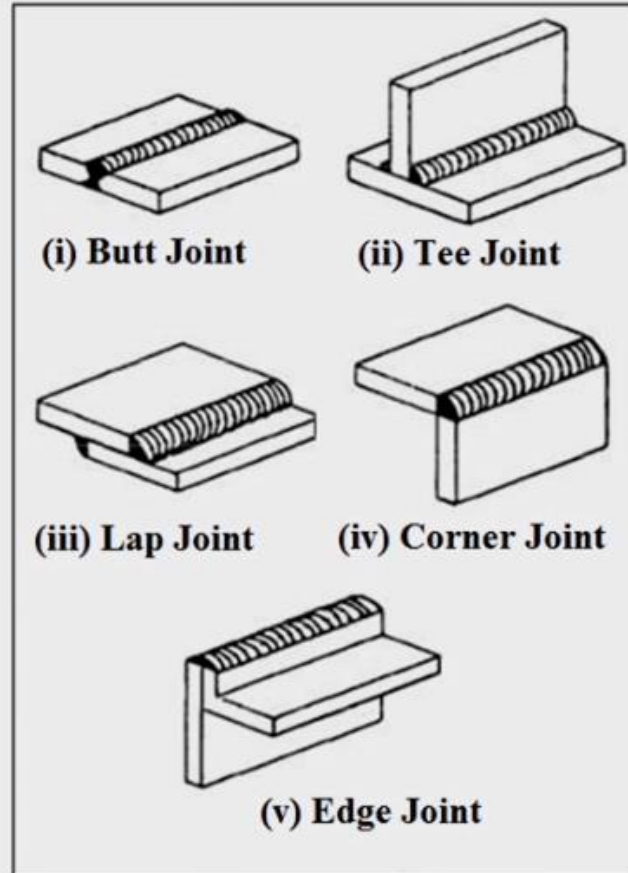
i) Butt joint

ii) Tee joint

iii) Lap joint

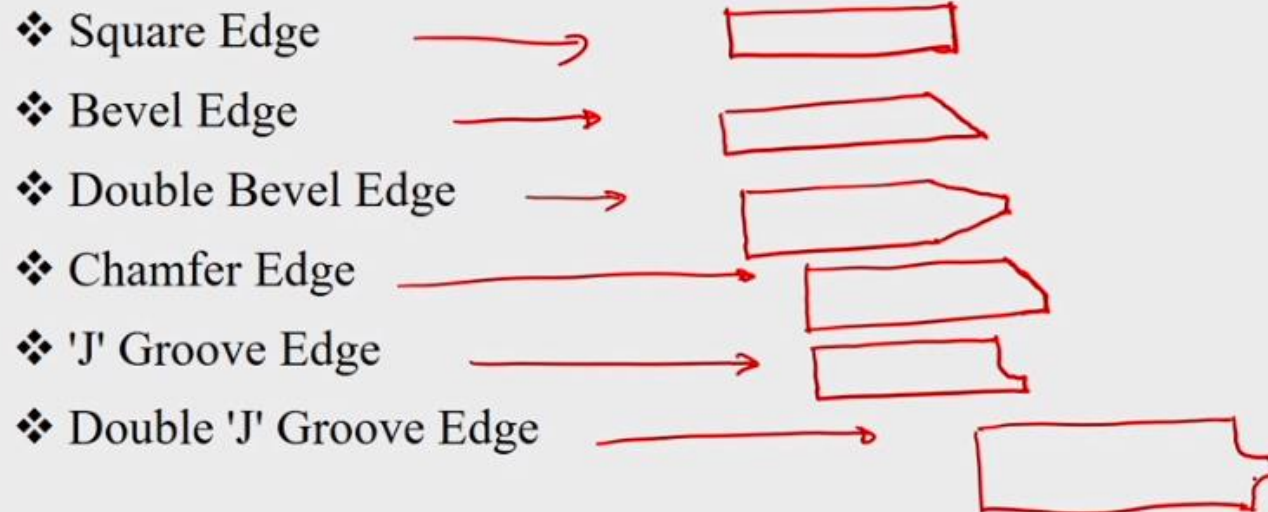
iv) Corner joint

v) Edge joint



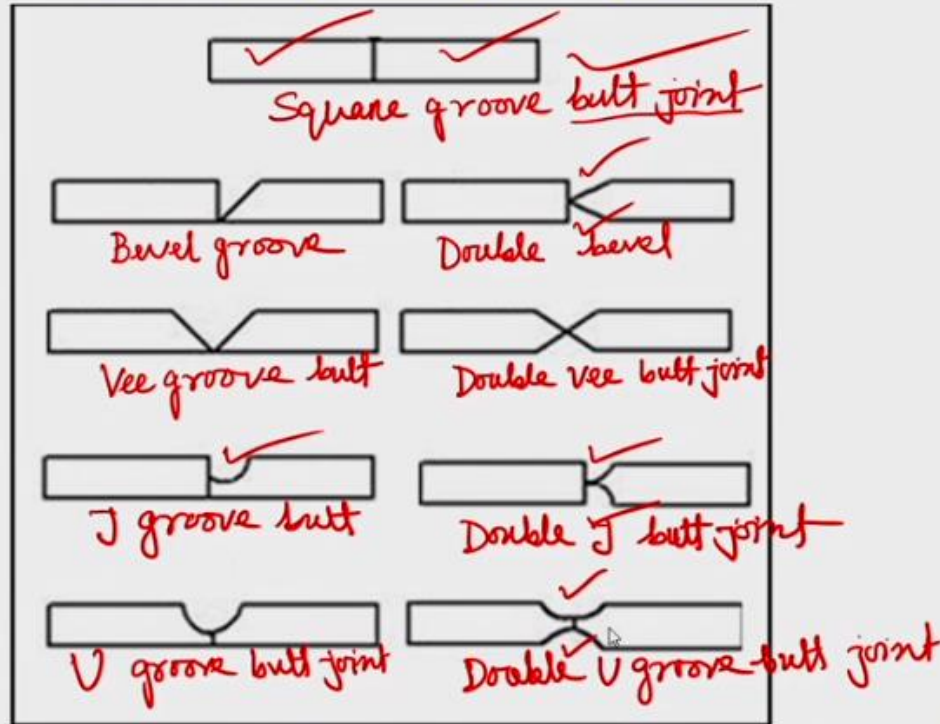
Edge Preparation of Weld Joints

- ❑ Edge preparation or edge shaping may be applied to each piece in the same way, or combinations of the joint preparations may be used.
- ✓ The edge preparation for welding the joint depends on the strength requirements and other design considerations.
- ✓ The most common edge preparations are:



Different Butt Joints and Edge Preparation

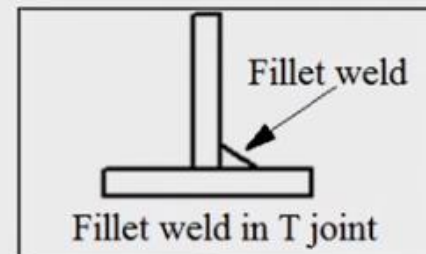
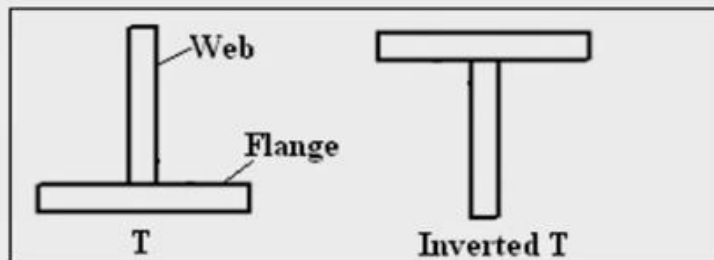
- ❑ A butt joint is formed when the pieces to be welded are laid side by side.
- ❑ This is one of the most widely used types of joint.



Note: The bevel groove and V grooves are easier to prepare and are used more than the J or U grooves.

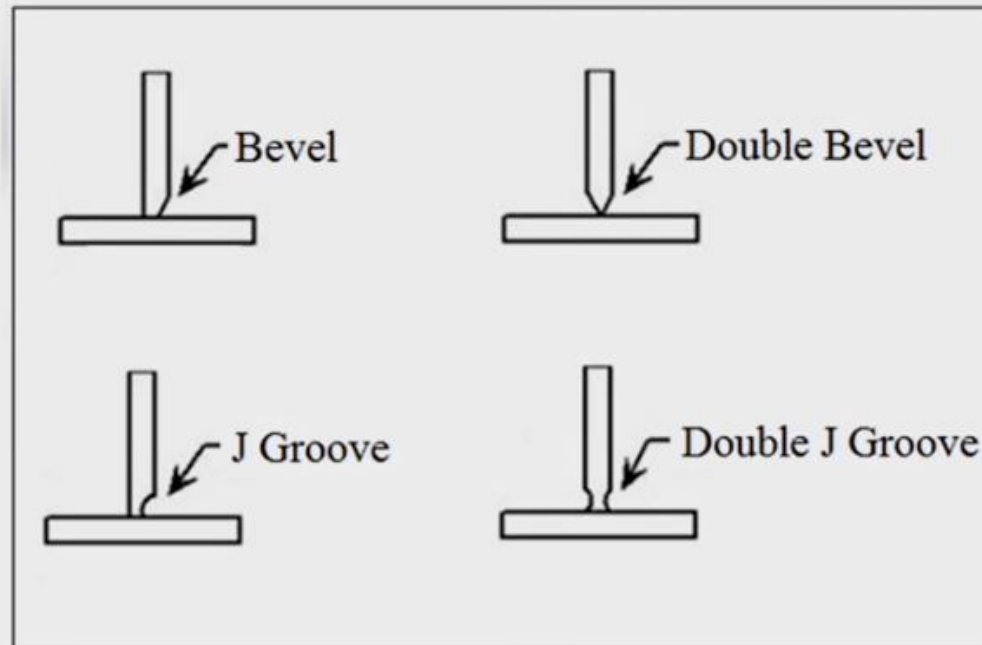
T Joints

- ❑ **T joint:** It is formed when one piece of metal to be welded is **placed vertically on another piece lying horizontally**, to form the shape of an inverted 'T'.
- ❑ The vertical member is usually placed at an angle of 90 deg. to the horizontal member.
- ❑ The most widely used types of weld applied to a T joint is the **fillet weld**. The weld resembles a 'Triangle' when viewed from the side.



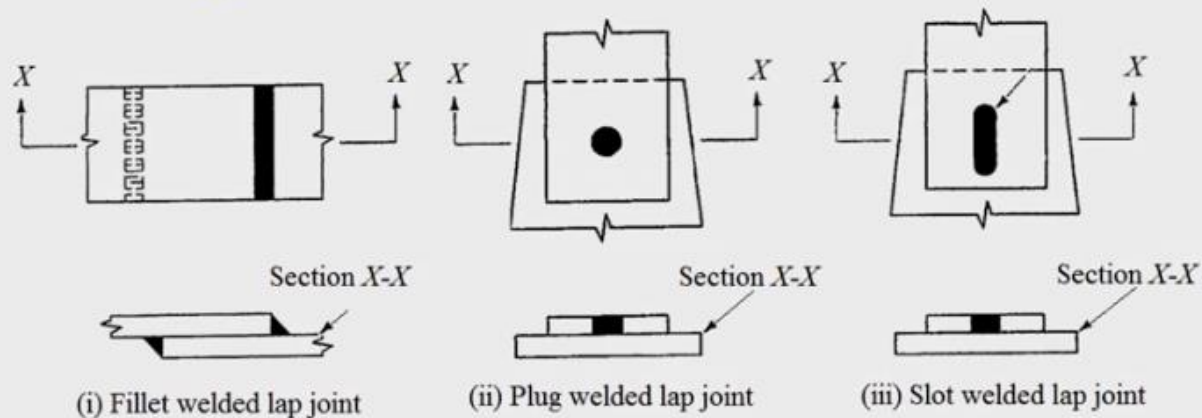
Different T Joints and Edge Preparation (*cont.*)

- ❑ The tee joint's vertical piece can also be prepared using a Bevel or J shape.



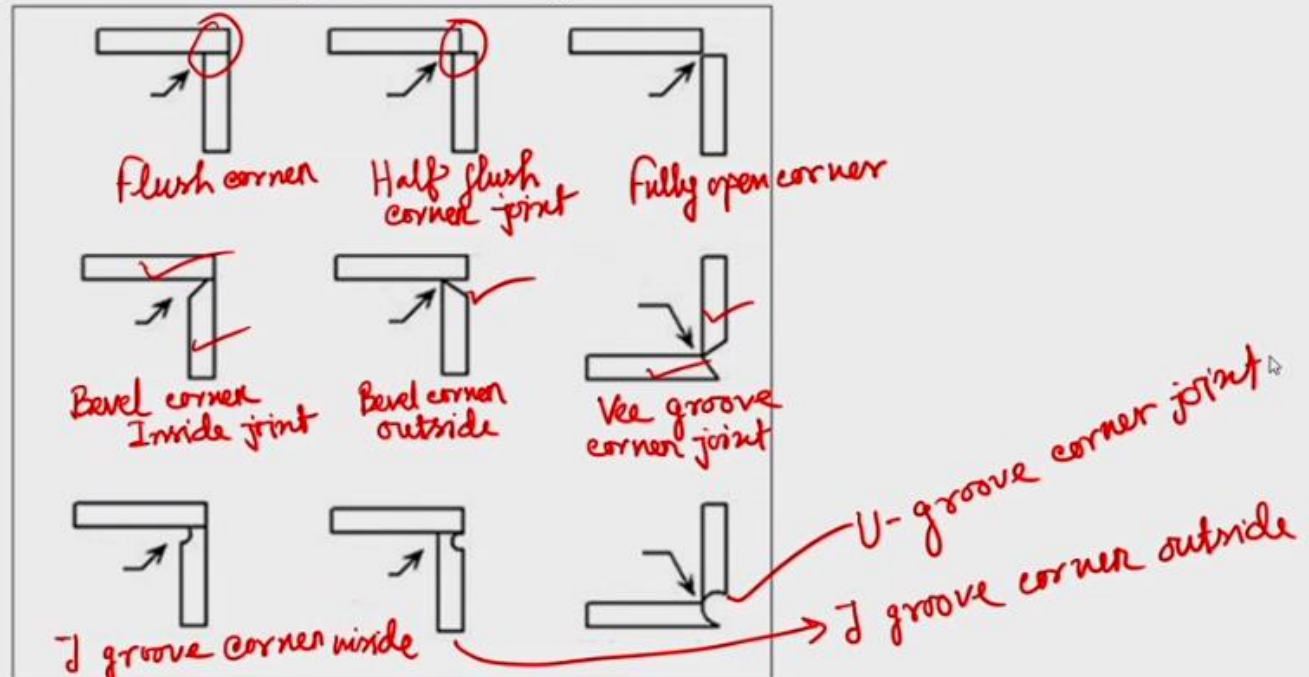
Different Lap Joints

- ❑ A Lap joint is formed when one piece to be welded is **laid down** and another piece is **overlapped** to form an edge for fillet welding or an area to allow plug or slot welding.
- ❖ A **plug weld** is made by welding holes evenly spaced across the length of one or both sides of the joint.
- ❖ A **slot weld** is similar except slots are made instead of holes.
- ❖ Other welds may also be applied to laps such as seam or projection welding.



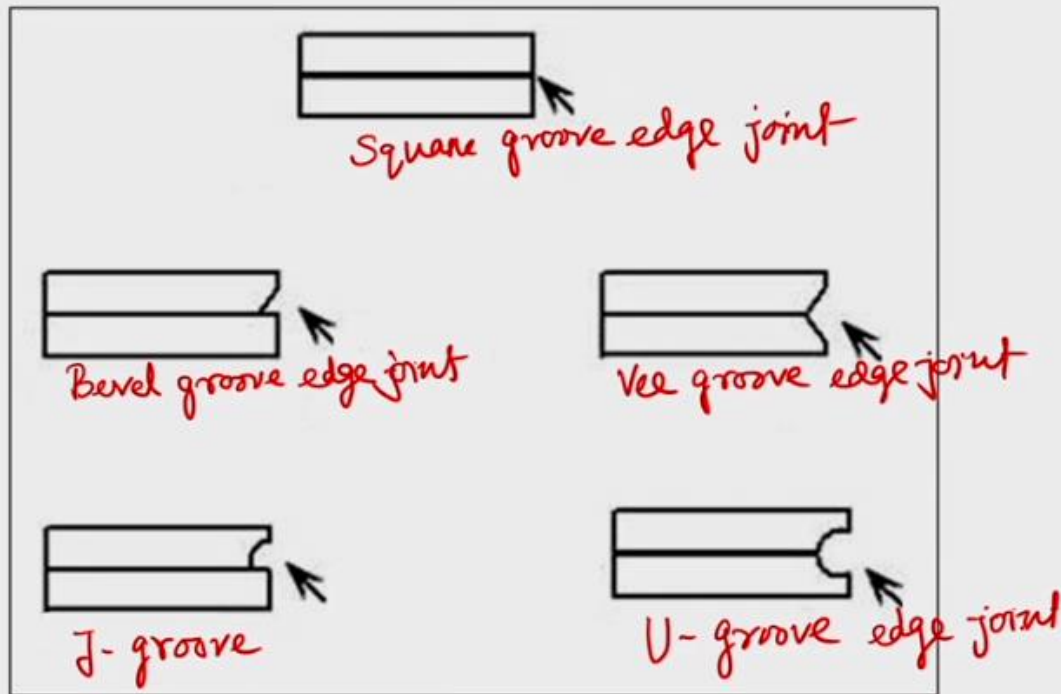
Different Corner Joints and Edge Preparation

- ❑ **Corner Joint:** A corner joint is formed by placing one piece to be welded on the other so that a corner is formed. It may be **Flush;** **Half Open;** or **Fully Open**.
- ❖ For proper penetration and strength, an edge preparation may be applied to one, or both of the pieces of the joint.



Different Edge Joints and Edge Preparation

- ❑ An Edge joint is formed when the two edges of the pieces to be welded come together.
- ❖ This joint is not as widely used as like the other type of joints.



Shape & name of different weld



Butt



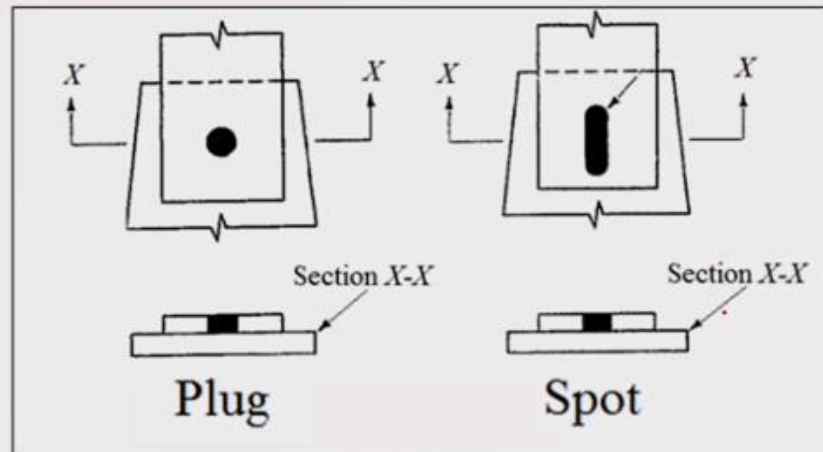
Fillet



Compound



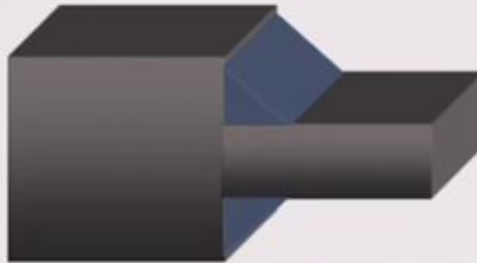
Edge



Shape & name of different weld *cont.*



(i) welded butt joint



(ii) welded butt joint

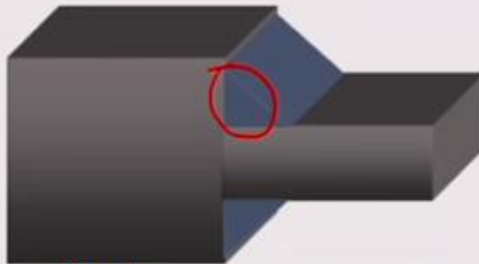


(iii) welded butt joint

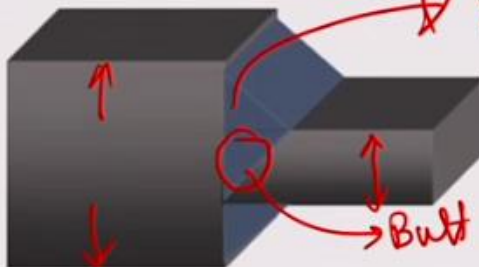
Shape & name of different weld *cont.*



(i) Butt welded butt joint



(ii) Fillet welded butt joint



(iii) Compound welded butt joint

Fillet weld

Butt weld

Shape & name of different weld *cont.*



(i) _____ welded T joint



(ii) _____ welded T joint

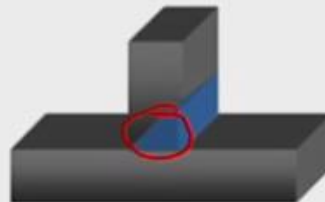


(iii) _____ welded T joint

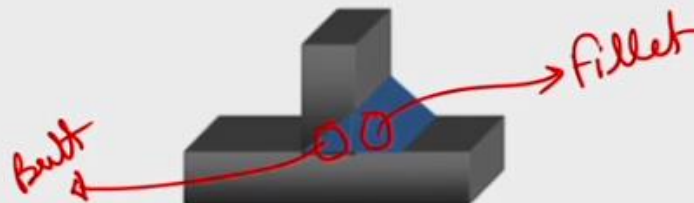
Shape & name of different weld *cont.*



(i) Fillet welded T joint



(ii) Butt welded T joint



(iii) Compound welded T joint

Shape & name of different weld *cont.*



(i) _____welded closed corner joint



(iv) _____welded open corner joint



(ii) _____welded closed corner joint



(v) _____welded open corner joint



(iii) _____welded closed corner joint



(vi) _____welded open corner joint

Shape & name of different weld *cont.*



(i) Butt welded closed corner joint



(iv) Fillet welded open corner joint



(ii) Inside fillet welded closed corner joint



(v) Outside fillet welded open corner joint



(iii) Compound welded closed corner joint



(vi) Double fillet welded open corner joint

Shape & name of different weld *cont.*



(i) **Fillet** welded lap joint



(ii) **Spot** welded lap joint



(iii) **Compound** welded lap joint

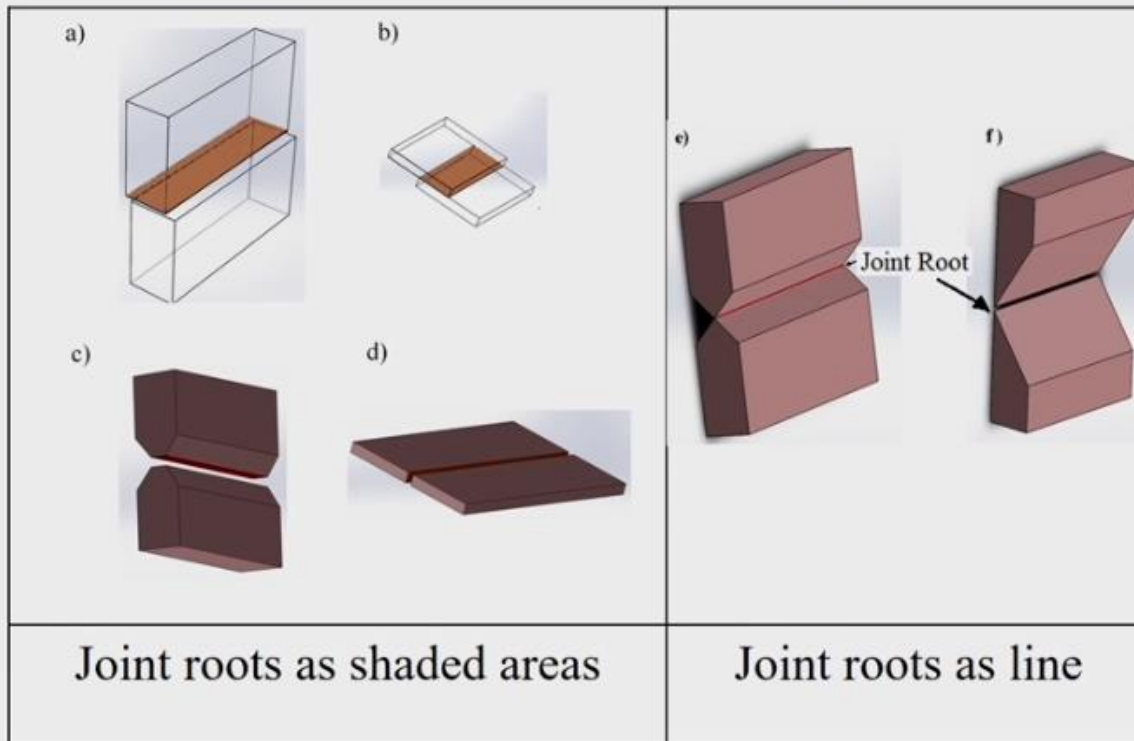
Parts of a weld

- Joint root
- Groove face
- Root face
- Root edge
- Root opening
- Bevel angle
- Groove angle
- Groove radius

Joint root

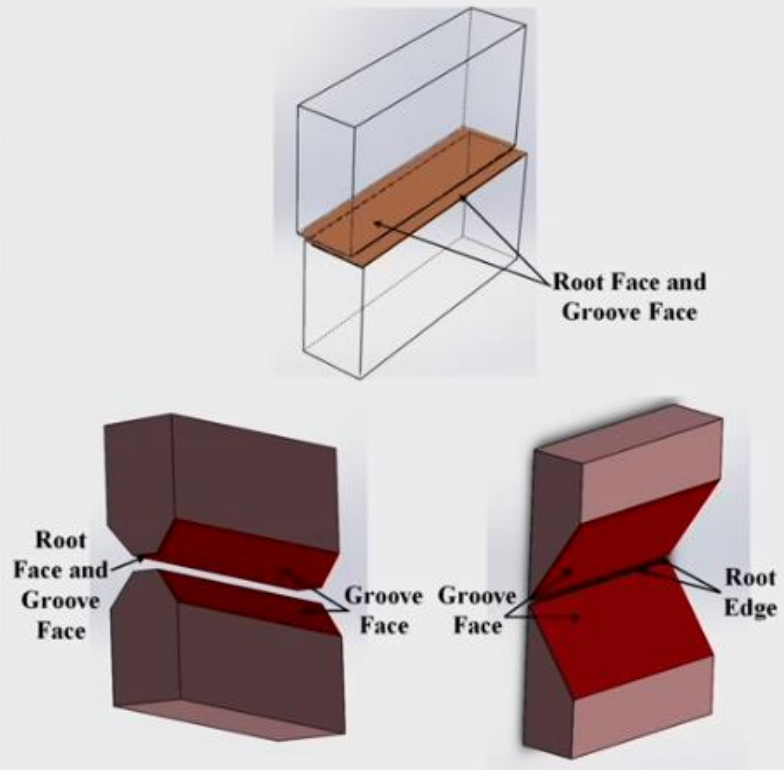
❑ **Joint root:** It is that portion of a joint to be welded where the **members are closest to each other.**

- The joint root may be either **an area, or a line.**



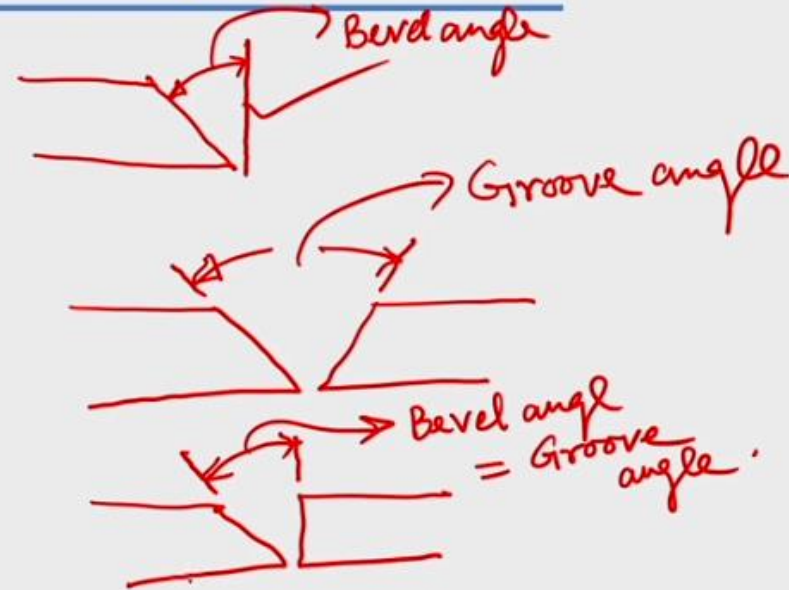
Groove Face, Root Face and Root Edge

- **Groove face:** It is all the surface of a member included in the groove.
- **Root face** is that portion of the groove face **within the joint root**.
- **Root edge:** It is a root face of zero width.



Bevel angle and Groove angle

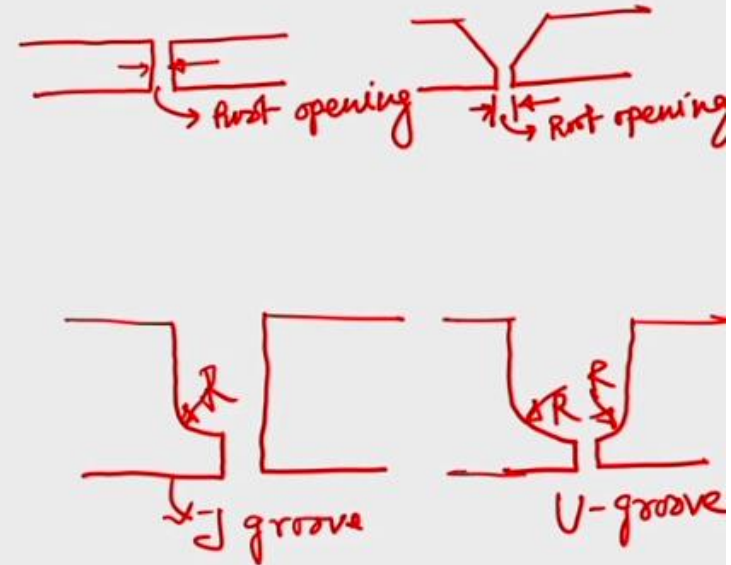
- **Bevel:** It is an angular edge preparation.
- **Bevel angle:** It is the angle between the bevel of a joint member and a plane perpendicular to the surface of the member.
- **Groove angle:** It the total included angle of the groove between two members.



Root opening and Groove Radius

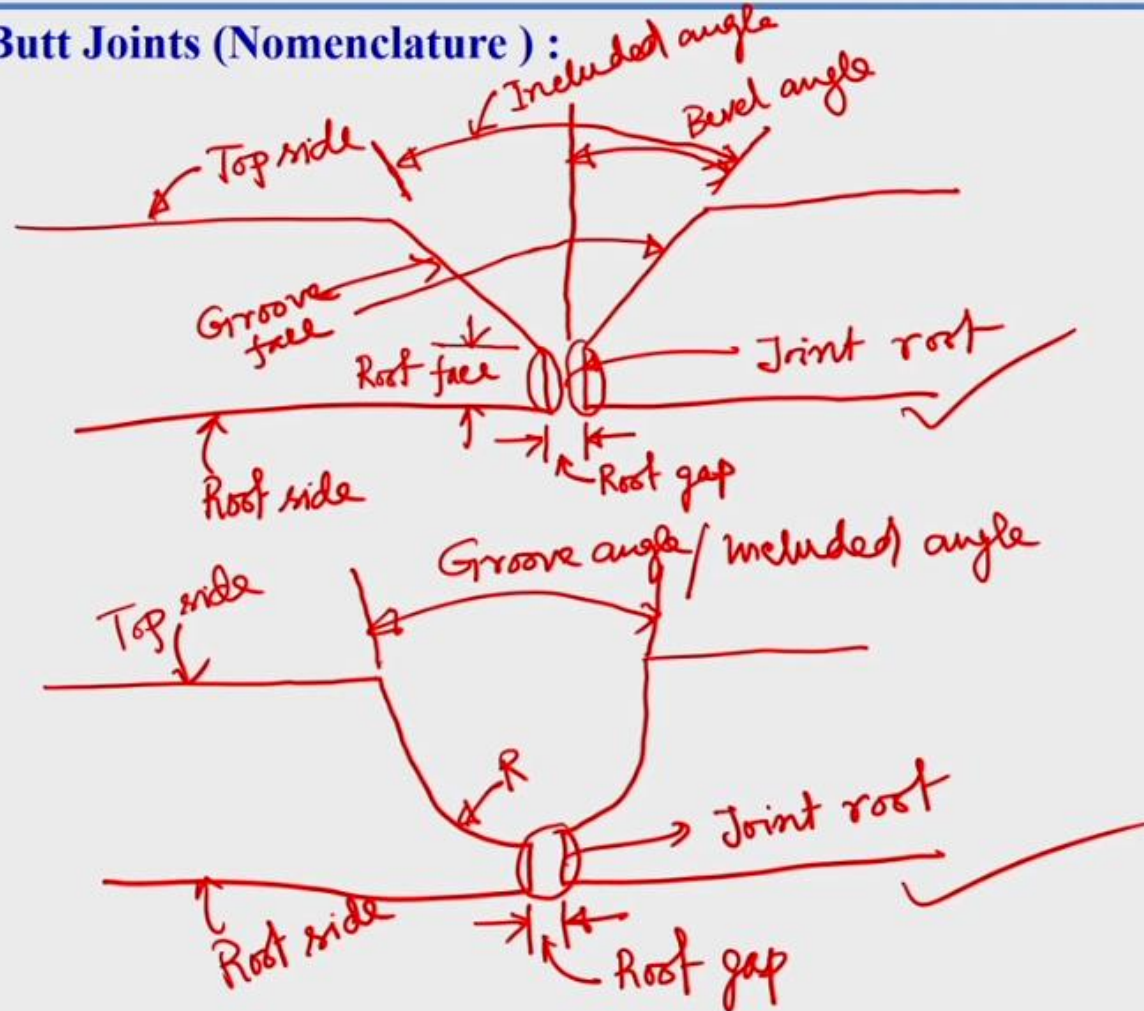
• **Root opening:** It is the separation between the work pieces at the joint root.

• **Groove radius:** It is an arc radius of a weld joint which applies only to J & U-groove welds. It is “the radius used to form the shape of a J- or U- groove weld.”



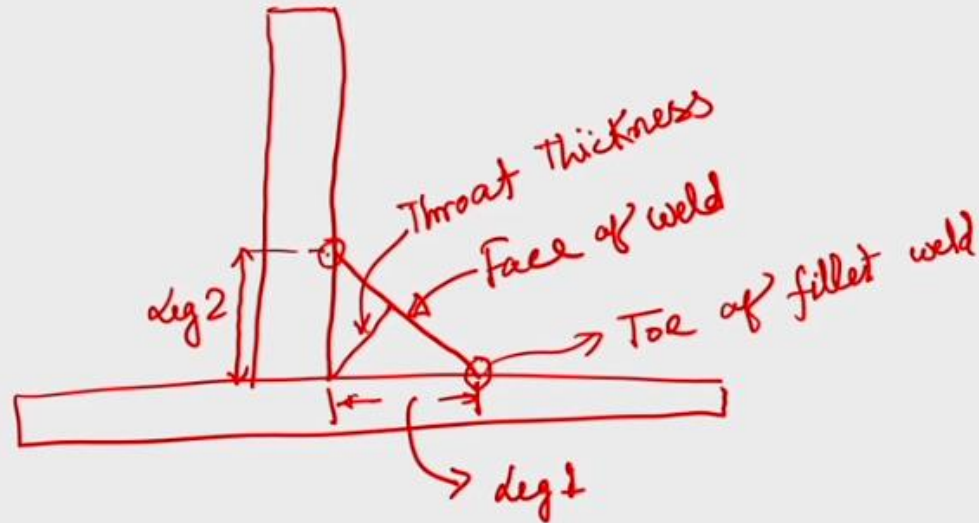
Detail Parts of Butt Joint

□ Parts of Butt Joints (Nomenclature) :



Different Parts of T Joint

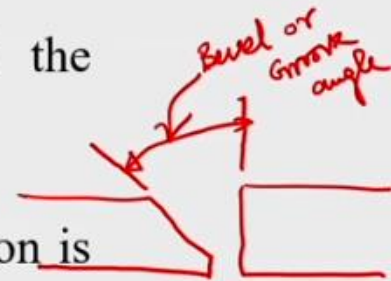
□ Parts of T-Joints (Nomenclature) :



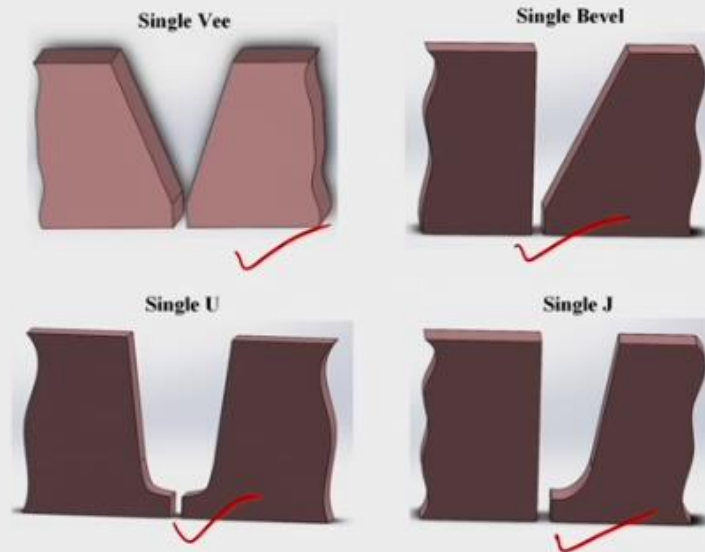
Extra necessary information

❑ Extra information is necessary to describe the exact joint design:

- For a single-bevel-groove-weld, the bevel angle and the groove angle are equal.
- In case of a J- or U- groove weld, the weld configuration is normally specified by both an angle and a radius.

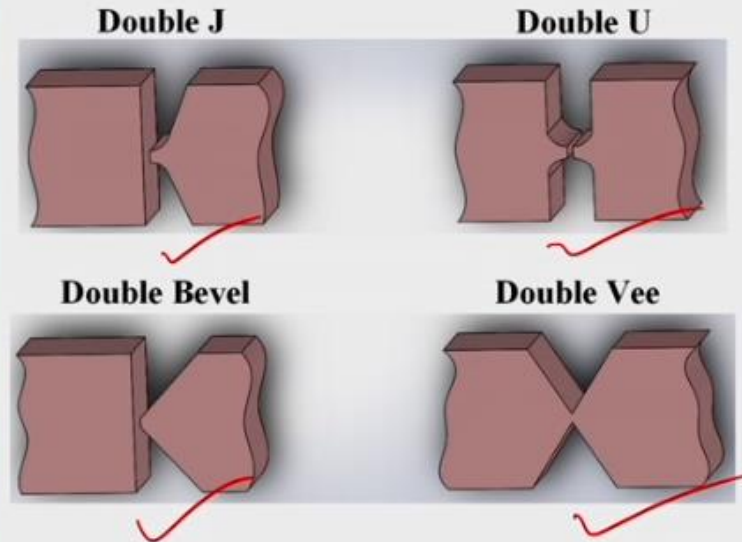


Single and double sided edge preparations



Note: Single sided preparations are normally made on **thinner materials**, or when access from both sides is restricted.

- ✓ A single-V butt joint is also more simpler to prepare.



Note: Double sided preparations are normally made on **thicker materials**, or when access from both sides is **unrestricted**.

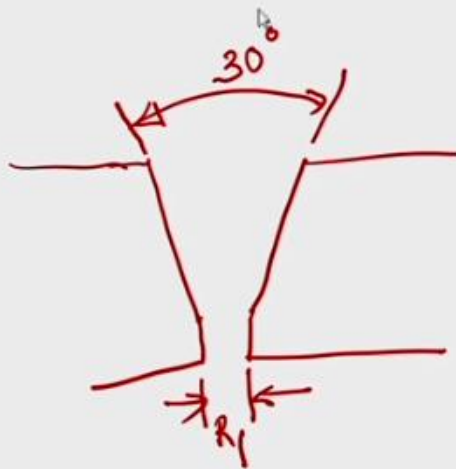
Single and double sided edge preparations



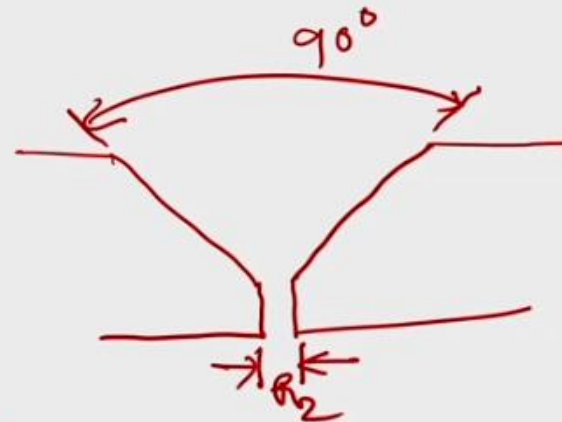
- ☐ **Note:** A double-V butt joint will cut down the amount of welding by almost one half in comparison to a single-V butt joint.
- ☐ A double-V butt joint reduces the amount of distortion.
- ☐ A double-V butt may require the joint **to be turned over** to complete the other side and access to both sides is required.

Root gap vs. the groove angle

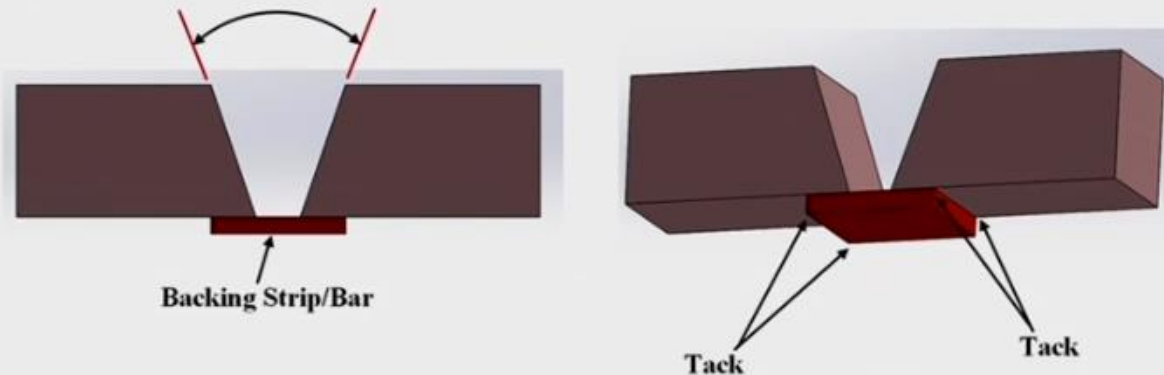
- ❖ The root gap is used for **electrode accessibility to the root** of the joint, the smaller the groove angle larger the root gap is required to achieve good penetration.



$$R_1 > R_2$$



Single-V with backing bar/strip



- A backing strip or backing bar is used mainly to support the root and to prevent burn through of weld joint.
- May be used for large root gaps and reduced groove angles.
- It can also allow for a feather edge to be used, no root face required.
- A backing strip usually forms part of the weld and a backing bar is usually removable.